# Naval Medical Research and Development News

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*Commanding Officer*
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# NMRC Hosts Visit from U.S. Global Malaria Coordinator, President's Malaria Initiative

SILVER SPRING, Md. - Rear Adm. (Ret.) Tim Ziemer, the U.S. Global Malaria Coordinator, President's Malaria Initiative, visited the Naval Medical Research Center (NMRC), November 29, for a brief on the current malaria vaccine research efforts and to tour the facility. He was interested in learning more about the malaria program at the laboratory.

Capt. John Sanders, NMRC commanding officer, provided a general overview of the NMRC enterprise with emphasis on the infectious diseases research efforts, specifically in the area of malaria.

"The NMRC malaria research program is at the forefront of malaria research worldwide," Sanders pointed out and added, "Researchers here have been investigating methods to control and conquer malaria for more than two decades and have made some exciting discoveries in the last few years."

Ziemer visited a laboratory focused on investigating the liver stage of infection as a vaccine target, spoke with a researcher about antigen discovery and another researcher on the humanized mouse model developed at NMRC. He also had the opportunity to visit the insectary and hear about clinical immunology and current malaria

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*Dr. Xiaoyan "Cathy" Zou, staff scientist from the Henry Jackson Foundation, discusses research on malaria with Rear Adm. (Ret.) Tim Ziemer, the U.S. Global Malaria Coordinator, President, Malaria Initiative.*

# NAMRU-3 Partners to Build Medical Capacity in Afghanistan

*From NAMRU-3 Public Affairs*

CAIRO - As part of U.S. Naval Medical Research Unit No. 3's (NAMRU-3) ongoing efforts to build medical capacity with Ministry of Health laboratories in several countries, NAMRU-3 is partnering with the Defense Threat Reduction Agency (DTRA) Cooperative Biological Engagement Program (CBEP) in Afghanistan. This collaboration enhances the efficiency and synergy in the U.S. government's biodefense and disease surveillance efforts.

According to NAMRU-3 Commanding Officer, Capt. Buhari Oyofo, "NAMRU-3 has developed training materials tailored to the Afghanis with special consideration of cultural practices."

NAMRU-3 has been involved in developing Afghanistan's public health capacity since 2006.

NAMRU-3's initial engagement was focused on the Ministry of Public Health (MoPH) and the Afghan Public Health Institute. NAMRU-3 assessed the capacity and capability of laboratory, staff and laboratory support facilities. First focusing on the Central Public Health Laboratory (CPHL) in Kabul, the program later included additional facilities in Kabul with plans for other regions of Afghanistan.



*NAMRU-3 hosted nine Afghan trainees from the Central PublicHealth Laboratory in Kabul for a bacteriology training workshop.*

Other efforts have been focused on assessing diagnostic capabilities; determining critical needs for supplies or equipment such as refrigerators, autoclaves or serology kits; evaluating existing training and licensing programs; and determining the need and MoPH interest in developing a train-the-trainer program. In 2011 NAMRU-3 provided training for 160 Afghan scientists and technicians on laboratory operations, diagnostic procedures, and ethics in research and management with respect to activity involving U.S. select agents. With DTRA support, a comprehensive training plan was developed for 2012 based on needs and gaps identified by NAMRU-3 laboratory assessments. NAMRU-3 researchers developed nine modules on parasitology, bacteriology, bioscience facility management, clinical epidemiology, miomedical equipment repair, laboratory quality management system, serology, molecular biology and virology.

***In 2011 NAMRU-3 provided training for 160 Afghan scientists and technicians on laboratory operations, diagnostic procedures, and ethics in research and management with respect to activity involving U.S. select agents...a comprehensive training plan was developed for 2012 based on needs and gaps identified by NAMRU-3 laboratory assessments.***

Information and sample flow to/from laboratories, including mechanisms and corresponding gaps in sample transport and information flow, were determined. In coordination with DTRA, NAMRU-3 provided needed supplies and training to fill the gaps.

NAMRU-3 established five hospital laboratories as well as virology, bacteriology and serology laboratories within the CPHL. They have provided training for various diagnostic laboratories and through implementation of an acute febrile illness/diarrhea study.

NAMRU-3 has also conducted several workshops to train laboratory and administrative staff on proper laboratory procedures, establish inventory for supplies, institute quality control procedures and standard operating procedures, purchase reliable supplies, and develop national laboratory biosafety and laboratory quality control plans.

# Presidential Memorandum - Accelerating Technology Transfer

*By Todd Ponzio, NMRC Technology Transfer Office*

SILVER SPRING, Md. - The "Presidential Memorandum - Accelerating Technology Transfer and Commercialization of Federal Research in Support of High-Growth Businesses" was released toward the end of last year. This memorandum's objectives directed Federal agencies to "[1] establish goals and measure performance, [2] streamline administrative processes, and [3] facilitate local and regional partnerships in order to accelerate technology transfer and support private sector commercialization."

'Technology transfer and commercialization' is a phrase used by the business side of research and development that manages the wide range of discoveries, inventions and other intellectual property within the Naval Medical Research Center (NMRC) enterprise. The goal is bringing these discoveries to market for the benefit of the warfighter.

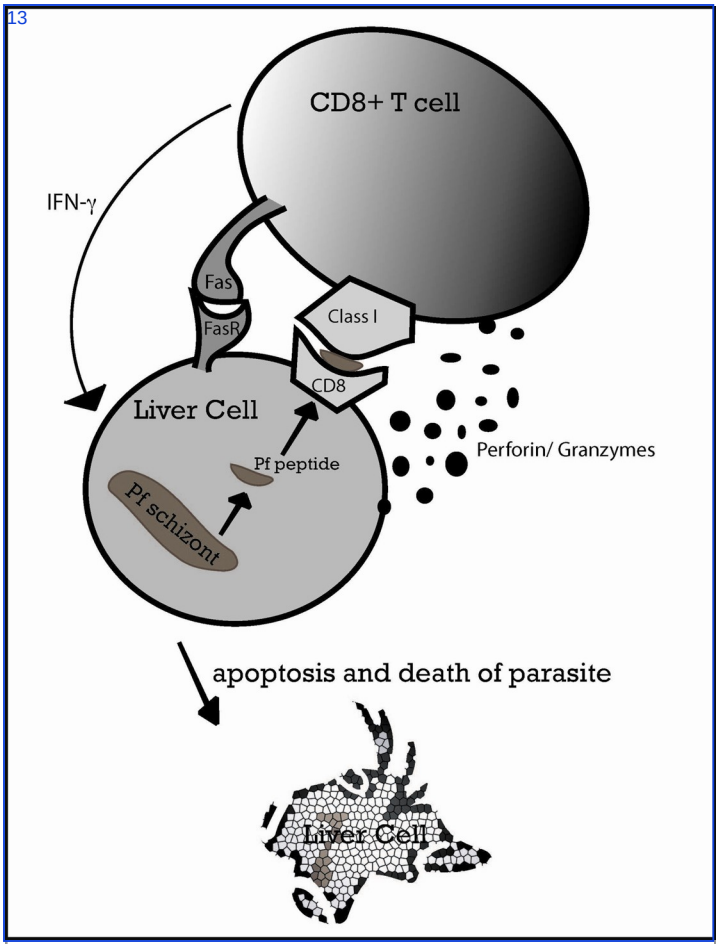
The NMRC enterprise excels in facilitating local and regional partnerships due to the inventiveness and creativity of our research scientists and physicians. Most valuable biomedical developments begin in a laboratory, but the value to the warfighter requires moving that discovery through all the business steps, ultimately to manufacturing and distribution. This requires the establishment of appropriate technology transfer agreements. NMRC's technology transfer collaborations leverage research capabilities found in the public and private sectors to stretch research dollars and accomplish the mission of supporting the health and readiness of the military's men and women in uniform.

For example, instead of a company having to reinvent something that is already available in a Navy laboratory, or vice versa, a partnership can capitalize on each collaborator's expertise toward achieving a shared goal. Every time an invention from one of our laboratories is commercialized into an accessible product, the U.S. taxpayer sees a tangible return on their investment and the Department of Defense has a product that will enhance the readiness of military personnel.

**Cooperative Research and Development Agreements (CRADAs) are key business vehicles among Navy biomedical scientists. This past quarter, the NMRC enterprise executed just over twenty CRADAs and reviewed an additional fourteen CRADAs for the Naval hospitals in Portsmouth and San Diego.**

One notable collaboration is being spearheaded by Cmdr. Jonathan Forsberg (NMRC, Operational and Undersea Medicine Directorate), who is working with a company to explore a novel mode of anchoring prosthetics. This project could produce important results for amputees. Another collaboration being led by Lt. Roxanne Burrus (U.S. Naval Medical Research Unit No. 6, Lima, Peru) involves Duke University and focuses on evaluating the effects of changing demography and land use on malaria transmission. This is an important issue in light of the prevalence of malaria in developing countries and is important to the health of deployed warfighters. An additional malaria-focused collaboration is being headed by Lt. R. Vince Gerbasi (NMRC, Infectious Diseases Directorate), who is using mass spectrometry to identify novel antigens to be used as potential vaccine candidates. Lastly, Dr Bjorn Song (Naval Medical Research Unit-San Antonio) is leading a collaboration focused on exploring the use of a synthetic oxygen-carrying fluid to reduce tissue damage occurring in response to hemorrhagic shock.

All of these collaborations have significant relevance to the military, but there is also considerable potential for these new sources of information and technologies to benefit the general population. Through the appropriate leveraging of resources through CRADAs and the commercialization of Navy Medicine inventions through patent licensing agreements, the NMRC enterprise excels at technology transfer. This is a key part of what the Presidential Memorandum was all about.




*A graphic illustration depicting Lt. R. Vince Gerbasi's effort to use mass spectrometry to identify novel antigens to be used as potential malaria vaccine candidates.*